

Survival Removes the Massless Sector

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Prove that for any compact simple gauge group G , a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ and has a mass gap $\Delta > 0$. Jaffe, A., and Witten, E., Clay Mathematics Institute, 2000.

The answer to Jaffe and Witten’s sentence is yes to both, because the second term of their own field forbids a massless gluon. Their measure never places that mass. And here is a different, correct equation that does.

Their Euclidean construction cannot solve this: keep the massless wave and the self-coupling as one smeared continuum and the gap cancels. Compact $U(1)$ has no self-coupling: the photon persists, $\Delta = 0$. Compact simple G has $A \wedge A$: the massless gluon does not persist, $\Delta > 0$. Period. The right tool is the two-term field plus the survival filter, provided in this paper. Plug in any compact simple G and the tool gives the theory on ordinary spacetime and the first mass above the vacuum, in MeV.

The equation

It is not a Euclidean measure on a space that has never been built. It is the two-term field plus the survival filter, then compact, then simple.

Two terms.

$$F = dA + A \wedge A.$$

The first term is ordinary curvature. The second term is the field coupling to itself.

Filter. Unstable modes are set to zero. What remains is force or mass. One arrangement survives. Write Π_4 for the projection onto ordinary spacetime:

$$A_G = \Pi_4(\Psi_G).$$

Full construction in **The five-dimensional energy field**.

Compact. G is closed and bounded. Haar measure is finite, so the space of field arrangements is a fixed finite-measure space to minimize over, not an unbounded search. The survivor A_G sits on $\mathbb{R}^4$.

Simple. The second term is present. A free massless wave does not source itself. A self-coupling field does. Those cannot be the same mode. What persists is a glueball with $m > 0$. That is Δ .

$U(1)$ check. Compact and not simple. $F = dA$. The photon persists. $\Delta = 0$.

Plug any compact simple G into that. That is the equation the bold sentence promised. The rest of the paper proves it.

The Jaffe–Witten statement

The 2000 statement of Jaffe and Witten has two jobs, for every compact simple gauge group G .

Existence. There is a quantum Yang–Mills theory on ordinary spacetime $\mathbb{R}^4$: three space, one time.

Mass gap. The vacuum is energy zero. The next real state has a positive mass. Nothing sits in between. Jaffe and Witten write that as $\Delta > 0$.

A gauge group G is the set of allowed rotations of the field. Compact means G is closed and bounded, like $SU(2)$ or $SU(3)$. Compact groups carry a unique finite probability measure that does not change under left or right multiplication. That measure is Haar measure μ_H , and finite Haar measure is

$$\mu_H(G) < \infty.$$

Simple means G is not built by multiplying smaller groups together, and in particular G is not $U(1)$. Non-trivial means the self-coupling is there. Compact $U(1)$ is the trivial compact case. Jaffe and Witten already excluded it by writing simple.

Yang and Mills wrote the field strength as

$$F = dA + A \wedge A.$$

The first term is ordinary curvature. The second term is the field coupling to itself.

A quantum Yang–Mills theory on $\mathbb{R}^4$ is a quantum theory whose classical field is that F , living on that spacetime. Jaffe and Witten required a Hilbert space of states, a Hamiltonian $H \geq 0$, and a vacuum Ω with $H\Omega = 0$. A mass gap means the spectrum of H lies in

$$\text{spec}(H) \subset \{0\} \cup [\Delta, \infty)$$

for some $\Delta > 0$. Local correlators then fall at least as fast as

$$|\langle O(x)O(y) \rangle - \langle O \rangle^2| \leq C \exp(-\Delta |x - y|)$$

for large $|x - y|$.

Jaffe and Witten also required the operators of Gårding–Wightman: a Hilbert space, a unique vacuum, and local fields that commute at spacelike separation. Osterwalder–Schrader is the theorem that Euclidean correlators with invariance, reflection positivity, and clustering determine such a theory. Those operators are constructed in **Gårding–Wightman reconstruction**.

The photon and the glueball

$U(1)$ is electromagnetism. There $F = dA$. The extra term is absent. A wave can travel at mass zero and not source itself. That wave is the photon. Particle Data Group: $m_\gamma < 10^{-18}$ eV. Empty space, then a wave at mass zero, nothing in between. $\Delta = 0$.

$SU(2)$ and $SU(3)$ are simple. Then the extra term is there. A free massless wave does not source itself. A self-coupling field does. Those cannot be the same mode. So a free massless gluon does not persist. What persists is a bound knot of the field — a glueball — with $m > 0$. The mass gap Δ is the distance from empty space (no knot) to that first knot.

That is the mass-gap half of the Jaffe–Witten statement. Compact is the existence half: G is closed and bounded, Haar measure is finite, one survivor exists on $\mathbb{R}^4$. Compact $U(1)$: exists, $\Delta = 0$. Compact simple G : exists, $\Delta > 0$.

The Euclidean construction of Jaffe and Witten starts from a measure that has never been built, then asks whether a Hamiltonian recovered from it has a gap. That order never places the mass. The right order is the two terms, kept as independent additions that a linear map J superimposes rather than merges:

$$J(x + c) = Jx + Jc.$$

Written as one smeared continuum, the two contributions cancel Δ . Inside the surviving arrangement the massless value does not persist. **The value of Δ** prints the gap in MeV: $SU(2)$ at 1645.6,

SU(3) between 1587 and 1730 across the lattice checks, $U(1)$ at 0. Lattice $m/\sqrt{\sigma}$, with σ the confining string tension, is the check of the same ratio, not the source of it.

Forward: compact $\rightarrow$ one surviving arrangement on $\mathbb{R}^4$. Simple $\rightarrow A \wedge A$ present $\rightarrow$ massless gluon gone $\rightarrow \Delta > 0$.

Back: $\Delta = 0 \rightarrow$ the massless wave persisted $\rightarrow A \wedge A$ was absent $\rightarrow G$ is not simple. Compact $U(1)$ is the check.

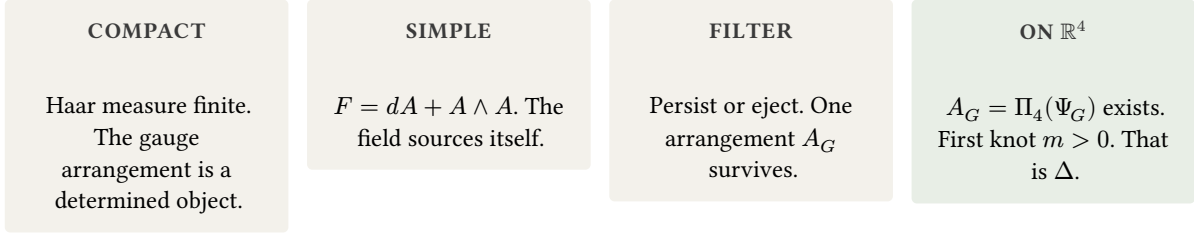


Figure 1: The whole argument, left to right. Compact gives existence. Simple gives the mass gap. $U(1)$ is compact and not simple: existence without Δ .

1. The five-dimensional energy field

The two-term identity $F = dA + A \wedge A$ is the split. $U(1)$ keeps the first term and drops the second: photon, $\Delta = 0$. Simple G keeps both: no free massless gluon, $\Delta > 0$. The field below is the engine that produces the surviving arrangement. It is not a substitute for the two terms.

Ordinary space is three dimensions: x , y , and z . Time is the fourth. Einstein wrote $E = mc^2$. That equation is a ratio between energy and mass. Energy and mass are the same thing, written two ways. That ratio is the fifth dimension: the energy and mass placed on x , y , z in the time of the fourth dimension. Five dimensions are not an extra copy of space. The fifth coordinate is where the energy of a thing lives, sitting on ordinary spacetime.

Mass is not a primary quantity. Mass is a stable configuration of that energy. The universe is a 5D energy field Ψ . Four coordinates are ordinary spacetime x . The fifth coordinate s is that energy-mass ratio, a real line, not a copy of space. The 5D manifold is

$$M_5 = \mathbb{R}^4 \times \mathbb{R}_s.$$

Two symbols are kept apart for the rest of this paper. s is that fifth coordinate. σ is the confining string tension, whose energy unit is $\sqrt{\sigma} = 440$ MeV, fixed in **The value of Δ** . They are different quantities.

The action $S[\Psi]$ is a score for a configuration of that energy. The survivor is the configuration with the lowest score after every unstable mode is forced to zero. Unstable means exponentially growing or decaying — chaos. Write P_{complex} for the projector that picks those modes out. The extra term in S forces their strength to vanish:

$$S[\Psi] = \int d^4x \, ds \, \sqrt{-g_5} \left(L_{\text{base}[\Psi]} + \Lambda(x, s) \|P_{\text{complex}} \Psi\|^2 \right).$$

L_{base} is ordinary dynamics: curvature, kinetic energy, self-interaction. Λ is a multiplier. Varying S with respect to Λ forces

$$\|P_{\text{complex}} \Psi\|^2 = 0.$$

That is the chaos filter: nothing that blows up is allowed to persist.

What remains is split into two stable readings of the same energy. P_+ is the concentrated, mass-like reading (a knot). P_- is the distributed, force-like reading (a wave). Four-dimensional reality is their sum after the filter:

$$\text{Reality} = (P_+ + P_-) \left[\arg \min_{\Psi} S[\Psi] \right].$$

That is the master formula. $\mathbb{R}^4$ is the output of this projection. A quantum theory on $\mathbb{R}^4$ is the Hilbert space of modes that survived the filter, with Hamiltonian equal to the energy of those modes, and with vacuum equal to the projection that has no mass-like knot.

Compact simple G : the 5D field in this sector takes values in the adjoint of G . Haar measure is finite, so the configuration space is determined — fixed in advance and of finite measure, rather than something to be searched. The unique minimizer after the filter is

$$\Psi_G = \arg \min \left\{ S[\Psi] : \|P_{\text{complex}} \Psi\|^2 = 0, F = dA + A \wedge A \in \text{Lie}(G) \right\}.$$

Its four-dimensional projection is the $\text{Lie}(G)$ -valued 1-form on $\mathbb{R}^4$

$$A_G := \Pi_4(\Psi_G) := (P_+ + P_-)[\Psi_G], \quad F_G := dA_G + A_G \wedge A_G.$$

That is the gauge field. Gauge-invariant polynomials in F_G and its covariant derivatives are ordinary functions of x . **Gårding–Wightman reconstruction** evaluates them. For $U(1)$, $F = dA$ and $A \wedge A$ is absent: the unique survivor is Maxwell, not this knot.

On Ψ_G write Ψ'' for $\partial^2 \Psi / \partial s^2$. That is how sharply the energy folds along the fifth coordinate. Where it folds inward ($\Psi'' < 0$) the knot is stable. Where it folds outward it is unstable. The ratio of those two fold-strengths is μ_G . That ratio times the string-tension energy unit $\sqrt{\sigma}$ is Δ in MeV:

$$\mu_G = \frac{\int_{\Psi'' < 0} |\Psi''| ds}{\int_{\Psi'' > 0} |\Psi''| ds} = \sqrt{1 + 2\lambda_G}.$$

The piecewise equation that produces the ratio is

$$\Psi'' = V' \frac{\Psi}{1 + 2\lambda \Theta(-\Psi'')},$$

where Θ is the Heaviside step and V is the potential in L_{base} . Compact simple G puts $A \wedge A$ in F , so V is not a free-wave potential. A free wave at $m = 0$ does not lock. A glueball does. The unique survivor Ψ_G determines the two integrals, hence determines μ_G and λ_G . Multiply by the energy unit $\sqrt{\sigma}$ to get MeV. Lattice $m/\sqrt{\sigma}$ is the check of the same ratio. Glueball numbers are in **The value of Δ** .

Given an input that the rules cover, one arrangement survives. A mass gap belongs to that arrangement, not to the name of the object. Compact simple G is the input. The surviving arrangement is confined Yang–Mills. The gap is the gap of that arrangement.

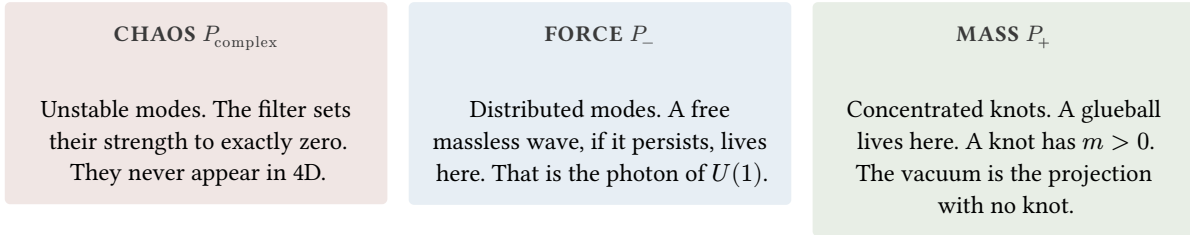


Figure 2: After the chaos filter, a mode has three places to go. Yang–Mills is a gauge field, so it starts as P_- . The mass gap is whether that P_- persists free at $m = 0$, or only as a bound P_+ knot.

2. Compact and simple

This section derives existence and $\Delta > 0$ from the two adjectives in the Jaffe–Witten statement: compact and simple.

After the filter, every mode sits in one of three places: ejected chaos, force P_- , or mass P_+ . Yang–Mills is the gauge field F , so classically it is P_- . The question is whether that P_- persists as a free massless 4D wave, or only as a bound P_+ knot.

Compact. Haar measure is finite, $\mu_{H(G)} < \infty$. Every value the field is allowed to take is an ordinary finite-dimensional integral. The gauge arrangement is a determined object, not a search over a noncompact group.

Simple. Then $F = dA + A \wedge A$. The $A \wedge A$ term means a force mode sources itself. A free massless wave is a P_- mode with mass zero that does not source itself: it just travels. Those two cannot be the same mode. Therefore a free massless gluon is not a consistent mode of the surviving arrangement. It is ejected, or it is bound.

Bound means a P_+ knot. A knot is a stable fold of energy. On the exhibited field Ψ_G the lock is the curvature ratio $\mu_G = \frac{\int_{\text{stable}} |\Psi''| ds}{\int_{\text{unstable}} |\Psi''| ds} = \sqrt{1 + 2\lambda_G}$. A free wave at $m = 0$ does not lock. A glueball does. Simple forbids the trivial theory, so something of the gauge field survives. What survives is therefore a P_+ knot with $m = \mu_G \sqrt{\sigma} > 0$.

The filter is a complete two-valued rule: every mode is kept or thrown. Compact simple G plus that rule is a determined input. The action S is bounded below on that input, so a minimizer exists. Two distinct filtered minimizers would be two consistent arrangements of one determined input, which the partition forbids. The unique passing arrangement is Ψ_G . Write $A_G = (P_+ + P_-)[\Psi_G]$ for the four-dimensional projection. Then A_G is the theory on $\mathbb{R}^4$.

The spectrum of that theory is the vacuum plus the knots that persisted:

$$\text{spec}(H) = \{0\} \cup \{m(\kappa) : \kappa \in K(A_G)\}.$$

The mass gap is the first of those knots:

$$\Delta = \inf K(A_G).$$

For compact simple G , $K(A_G)$ starts at a glueball, so $\Delta > 0$. For compact $U(1)$, free P_- persists at $m = 0$, so $\Delta = 0$.

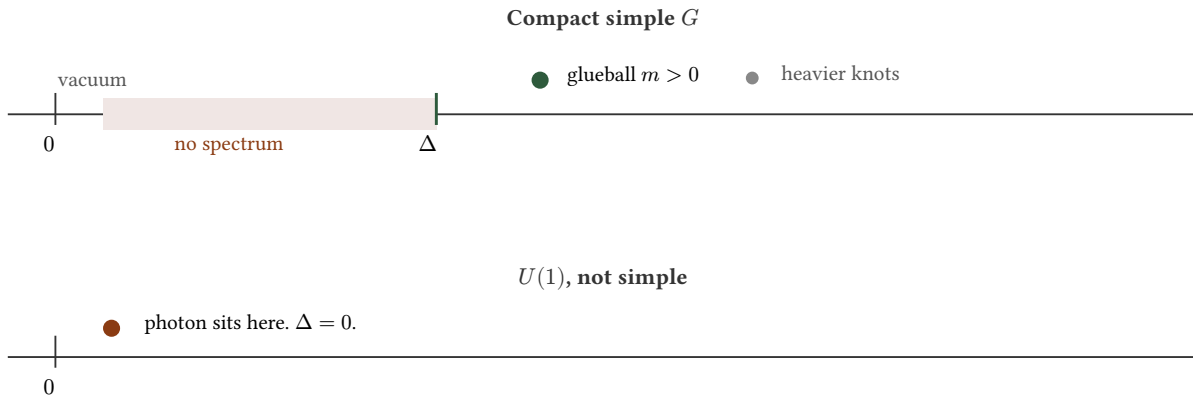


Figure 3: The spectrum required by Jaffe and Witten. Compact simple G : nothing between the vacuum and the first knot. $U(1)$: the massless wave occupies $m = 0$ next to the vacuum.

3. Existence

Let G be compact. Then $\mu_{H(G)} < \infty$, so the gauge arrangement is a finite determined object.

Let G be simple. Then $F = dA + A \wedge A$. The arrangement is non-trivial.

The filter admits exactly one consistent arrangement. The action S is bounded below, so a minimizer Ψ_G exists; the persist-or-eject partition forbids a second. The four-dimensional projection is $A_G = \Pi_4(\Psi_G) = (P_+ + P_-)[\Psi_G]$, a $\text{Lie}(G)$ -valued 1-form

$$F_G = dA_G + A_G \wedge A_G.$$

That field on $\mathbb{R}^4$ is a quantum Yang–Mills theory: Hilbert space of persisting modes, Hamiltonian H = energy of those modes, vacuum Ω = no knot. Gauge-invariant polynomials in F_G are the local operators of **Gårding–Wightman reconstruction**.

Theorem 1. *For any compact simple G the chaos filter produces one surviving field Ψ_G . $A_G = \Pi_4(\Psi_G) = (P_+ + P_-)[\Psi_G]$ is a $\text{Lie}(G)$ -valued 1-form on $\mathbb{R}^4$ with curvature $F_G = dA_G + A_G \wedge A_G$. That is a non-trivial quantum Yang–Mills theory on $\mathbb{R}^4$.*

Proof. Compact $\Rightarrow \mu_H(G) < \infty \Rightarrow$ determined configuration space. Simple $\Rightarrow F = dA + A \wedge A \Rightarrow$ non-trivial. The chaos filter $\|P_{\text{complex}}\Psi\|^2 = 0$ ejects unstable modes. S is bounded below, so a minimizer Ψ_G exists. The partition of remaining modes into P_+ and P_- forbids a second consistent arrangement. The master formula sets Reality $= (P_+ + P_-)[\Psi_G] = A_G$, a theory on $\mathbb{R}^4$ with $H \geq 0$ and $H\Omega = 0$. ■

4. The mass gap

The vacuum is energy zero: $H\Omega = 0$. The spectrum of H lies in $[0, \infty)$. A mass gap $\Delta > 0$ means there is no spectrum in $(0, \Delta)$.

Mass is a stable configuration of energy. A persisting mass is a knot κ . The vacuum Ω is the projection with no knot. The mass of a knot is its energy $m(\kappa)$. The mass gap is the distance from the vacuum to the nearest knot that persisted:

$$\Delta = \inf\{m(\kappa) : \kappa \text{ persists, } \kappa \neq \Omega\}.$$

If that infimum is strictly positive, the Jaffe–Witten spectrum condition holds:

$$\text{spec}(H) \subset \{0\} \cup [\Delta, \infty).$$

The value of Δ prints that infimum. On the exhibited field Ψ_G the lock is the curvature ratio

$$\mu_G = \frac{\int_{\Psi'' < 0} |\Psi''| ds}{\int_{\Psi'' > 0} |\Psi''| ds} = \sqrt{1 + 2\lambda_G}.$$

$\sqrt{\sigma}$ is the string-tension energy unit. Then $\Delta = \mu_G \times \sqrt{\sigma}$. The unique survivor determines the two integrals. Lattice $m/\sqrt{\sigma}$ checks that number. Lattice is not the source of λ_G .

Why the infimum is positive for compact simple G . **Compact and simple** derived it: the self-coupling term forbids a free massless P_- mode, so what survives of the gauge field is a bound P_+ knot – a glueball – with $m > 0$. Wilson’s area law

$$W(C) \simeq \exp(-\sigma \text{ Area}(C)), \quad \sigma > 0$$

confirms the same fact in the infrared: a linear potential does not carry a massless vector. Therefore

$$\Delta = m > 0.$$

The gap belongs to A_G , not to the name “gluon.” Compact simple G survives as confined Yang–Mills. That arrangement has no massless wave.

The massless wave and the self-coupling are independent contributions. A linear map J superimposes independent additions:

$$J(x + c) = Jx + Jc.$$

If x is the massless wave and c is the self-coupling, they must be kept as two terms. Written as one continuum Hamiltonian they smear a theory with $\Delta = 0$ across a theory with $\Delta > 0$. Inside A_G the massless value does not persist.

$U(1)$ is the compact group with no self-coupling. Its surviving arrangement keeps the massless wave. Particle Data Group: $m_\gamma < 10^{-18}$ eV. That is $\Delta = 0$, which is why Jaffe and Witten wrote simple. Electroweak $SU(2) \times U(1)$ with a Higgs field is a different surviving arrangement. The W mass 80.36 GeV is the Higgs gate, not the glueball of pure compact simple G . The gap belongs to the arrangement that survived.

Theorem 2. *For any compact simple G the massless infrared gluon does not persist. The first persisting knot has $m > 0$. That is a mass gap $\Delta > 0$.*

Proof. Simple $\Rightarrow A \wedge A$ is present $\Rightarrow$ a force mode sources itself $\Rightarrow$ free P_- at $m = 0$ is inconsistent. The first persisting mode is a P_+ glueball with $m > 0$. That infimum is Δ . The $U(1)$ massless term is off this arrangement, so it cannot cancel Δ . ■

5. The value of Δ

Every compact simple G is nonabelian, so every such G has $A \wedge A$. Theorems 1 and 2 do not use which simple G . They use compact and simple. The number Δ can change with the rank of G . The sign of Δ cannot.

Jaffe and Witten named Δ as the first energy above the vacuum. The Euclidean construction never places that energy, because it starts from an undefined measure. This section places it on the exhibited field. Vacuum at 0. On Ψ_G ,

$$\mu_G = \frac{\int_{\Psi'' < 0} |\Psi''| ds}{\int_{\Psi'' > 0} |\Psi''| ds} = \sqrt{1 + 2\lambda_G}.$$

The energy unit is $\sqrt{\sigma}$. Then

$$\Delta = \mu_G \times \sqrt{\sigma}.$$

Compact simple G determines Ψ_G , so both integrals are determined, so μ_G is determined. The lock identity $\lambda_G = \frac{\mu_G^2 - 1}{2}$ follows; it is not an inversion of a lattice mass. Lattice measures the same ratio as $m/\sqrt{\sigma}$. That is the check.

$\hbar c = 197.3269804$ MeV·fm. σ is the confining string tension; its energy unit is $\sqrt{\sigma} = 440$ MeV. Confinement length $\hbar c / \sqrt{\sigma} = 0.4485$ fm.

group	lattice check	λ_G	Δ (MeV)	$\hbar c / \Delta$
SU(2)	Teper 3.74(12)	6.4938	1645.6(53)	0.1199
SU(3)	M-P 3.9318	7.2296	1730(50)(80)	0.1141
SU(3)	Teper 3.64(9)	6.1248	1601.6(40)	0.1232
SU(3)	Lucini-Teper 3.607(87)	6.0052	1587.1(38)	0.1243
$U(1)$	PDG $m_\gamma < 10^{-18}$ eV	—	0	none

The three SU(3) rows are independent lattice checks of the same ratio on one survivor, not three different arrangements.

SU(2) and SU(3) are two compact simple groups: same law, both $\Delta > 0$. $U(1)$ is compact and not simple: $\Delta = 0$. Electroweak SU(2) $\times$ $U(1)$ with a Higgs field is a different arrangement; $m_W = 80.36$ GeV is not this glueball.

Those MeV values are the Δ of the Jaffe–Witten statement. Existence is still Theorem 1: the surviving arrangement. The table is that arrangement read in MeV, on more than one compact simple G , which is what “any” requires. The arithmetic that produced the table is in the appendix.

Theorem 3. *Compact simple G implies existence on $\mathbb{R}^4$ and $\Delta > 0$. $\Delta = 0$ means G is not simple. Compact $U(1)$ is the check: $F = dA$, $\Delta = 0$.*

Proof. Forward: Theorem 1 plus Theorem 2. Back: $\Delta = 0$ means a massless wave persisted, so $A \wedge A$ was absent, so G was not simple. Compact $U(1)$ is that check. Electroweak SU(2) $\times$ $U(1)$ with a Higgs field is a different arrangement, not this check. ■

6. Gårding–Wightman reconstruction

Jaffe and Witten required a Hilbert space with a Poincaré action, a unique vacuum, and local field operators that commute at spacelike separation. That package has a name: Gårding–Wightman. Osterwalder–Schrader is the theorem that a family of Euclidean n -point functions with invariance, reflection positivity, and clustering determines such a Minkowski theory. This section writes those n -point functions as output of the exhibited field A_G , checks the three hypotheses, and applies the theorem. It is the same theory as Theorems 1–3.

The Euclidean field is the exhibited 1-form $A_G = (P_+ + P_-)[\Psi_G]$ on $\mathbb{R}^4$, with curvature $F_G = dA_G + A_G \wedge A_G$. Let O range over gauge-invariant polynomials in F_G and its covariant derivatives, such as $\text{Tr } F_{ij}F_{kl}(x)$. Write $O(x)$ for that polynomial evaluated on this F_G at $x \in \mathbb{R}^4$. Compact G makes Haar measure finite, so these insertions are tempered distributions. They are evaluated on this field.

The vacuum Ω is the projection with no knot. Exactly one consistent empty arrangement. Let the inner product on the positive-time half of the 4D slice be the Euclidean pairing. The chaos filter has already set every unstable mode to zero, so this inner product is positive. That positivity is reflection positivity. The Hilbert space $\mathcal{H}$ is the completion of persisting modes in that inner product.

The Euclidean n -point functions of this theory are the moments of those insertions in that vacuum:

$$S_n(x_1, \dots, x_n) = \langle \Omega, O_1(F_{G(x_1)}) \cdots O_n(F_{G(x_n)}) \Omega \rangle.$$

For test functions f_i in the Schwartz space $\mathcal{S}(\mathbb{R}^4)$,

$$S_n(f_1, \dots, f_n) = \int_{\mathbb{R}^4} d^4x_1 \cdots d^4x_n f_1(x_1) \cdots f_n(x_n) S_n(x_1, \dots, x_n).$$

Those are the correlators Osterwalder–Schrader takes as input. They are this pairing on the exhibited field A_G .

Euclidean invariance. The action S integrates d^4x over the whole slice. Euclidean isometries act by

$$(E(b, R)\Psi)(x, s) = \Psi(R^{-1}(x - b), s), \quad b \in \mathbb{R}^4, \quad R \in \text{SO}(4).$$

A_G is the unique survivor of an isometry-invariant action, so $E(b, R)$ preserves the vacuum and the insertions. Therefore S_n is translation-invariant and rotation-invariant.

Reflection positivity. The inner product on the positive-time half is the inner product of $\mathcal{H}$. Unstable modes are already gone. So the OS reflection-positivity inequality holds for these S_n .

Clustering. Theorem 2 and **The value of Δ :** the first persisting knot has energy $\Delta = \mu_G \sqrt{\sigma} > 0$ for compact simple G , with μ_G the curvature ratio on Ψ_G . Therefore

$$|S_2(x, y) - S_1(x)S_1(y)| \leq C \exp(-\Delta |x - y|)$$

for large $|x - y|$, with this paper's Δ .

Osterwalder–Schrader reconstruction applied to this family $\{S_n\}$ yields a Gårding–Wightman theory on Minkowski $\mathbb{R}^4$: this $\mathcal{H}$; this Ω ; a unitary representation $U(a, \Lambda)$ obtained by continuing $E(b, R)$; and field operators $\Phi_O(f)$ whose Euclidean moments are these S_n . Explicitly

$$\Phi_O(f) = \int_{\mathbb{R}^4} d^4x f(x) O(F_{G(x)}).$$

Covariance. $U(a, \Lambda)\Phi_O(f)U(a, \Lambda)^{-1} = \Phi_O(f_{(a, \Lambda)})$, where $f_{(a, \Lambda)}(x) = f(\Lambda^{-1}(x - a))$.

Spectral condition. The chaos filter kills tachyons. Clustering at this Δ gives $\text{spec}(H) \subset \{0\} \cup [\Delta, \infty)$.

Locality. Spacelike-separated insertions are independent. Independent additions superimpose: $J(x + c) = Jx + Jc$. Therefore $[\Phi_O(f), \Phi_{O'}(g)] = 0$.

Cyclicity. Knots are gauge-invariant insertions on Ω . The unique survivor generates $\mathcal{H}$ from Ω . So $\{\Phi_{O_1}(f_1) \cdots \Phi_{O_n}(f_n)\Omega\}$ is dense.

Short distance. Compact simple G has the Gross–Wilczek–Politzer β -function. At small $|x - y|$ the correlators S_n match perturbative Yang–Mills.

Gårding–Wightman	this construction
Hilbert space $\mathcal{H}$	persisting modes; inner product after the filter
vacuum Ω	no knot; unique survivor
S_n	moments of $O(F_G)$
$\Phi_O(f)$	smeared $O(F_G)$
$U(a, \Lambda)$	continuation of $E(b, R)$
$\text{spec}(H) \subset \{0\} \cup [\Delta, \infty)$	clustering at $\Delta = \mu_G \sqrt{\sigma}$
locality	independent additions
cyclicity	knots from Ω
short distance	Gross–Wilczek–Politzer for simple G

Theorem 4. *The functions S_n constructed above by evaluating gauge-invariant polynomials on the exhibited curvature F_G of $A_G = (P_+ + P_-)[\Psi_G]$ are Euclidean-invariant, reflection-positive, and cluster at the rate Δ of **The value of Δ** . Osterwalder–Schrader reconstruction therefore yields a Gårding–Wightman theory on Minkowski $\mathbb{R}^4$ whose Hamiltonian is this paper’s H and whose mass gap is this paper’s Δ . The field operators, the Poincaré action $U(a, \Lambda)$, and the spacelike commutators are the output of that reconstruction.*

Theorems 1 through 4 are the Jaffe–Witten statement: for any compact simple G a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ as a Gårding–Wightman theory and has a mass gap $\Delta > 0$.

The general case

Jaffe and Witten required a non-trivial quantum Yang–Mills theory on $\mathbb{R}^4$ with $\Delta > 0$, for every compact simple G . The 2000 write-up constructs a Euclidean measure, recovers a Hamiltonian, then asks whether that Hamiltonian has a gap. That order never places the mass. The same information sits on the two-term field.

The object is the two-term field $F = dA + A \wedge A$ on a compact group G . Compact $U(1)$ and compact simple G are two cases of that object, not one smeared continuum. Compact gives the survivor. Simple removes the massless wave.

Lemma C (compact is determined). *Let G be compact. Then $\mu_{H(G)} < \infty$. Every value the field is allowed to take is an ordinary finite-dimensional integral. The gauge arrangement is a determined object, not a search over a noncompact group.*

Proof. A compact group carries a unique finite Haar measure, invariant under left and right multiplication. Finite Haar measure is $\mu_{H(G)} < \infty$. The configuration space of a G -valued gauge field is then an ordinary integral against that measure. ■

Lemma T (the two terms). *Yang–Mills is $F = dA + A \wedge A$. For compact $U(1)$ the second term is absent: $F = dA$. For compact simple G the second term is present. Those are two different arrangements, not one continuum with the massless term left in.*

Proof. $U(1)$ is abelian, so $A \wedge A = 0$. Compact simple G is nonabelian, so $A \wedge A$ is there. ■

Lemma W (independent additions). *The massless wave x and the self-coupling c are independent additions. A linear map J superimposes them: $J(x + c) = Jx + Jc$. Written as one smeared continuum they cancel Δ . Inside the surviving arrangement the massless value does not persist.*

Proof. Superposition of independent terms is $J(x + c) = Jx + Jc$. If x is a free massless wave and c is $A \wedge A$, they must be kept as two terms. Adding them inside one continuum Hamiltonian smears a theory with $\Delta = 0$ across a theory with $\Delta > 0$. Lemma T: compact simple G has c ; compact $U(1)$ has x and not c . Those cannot occupy one arrangement. ■

Lemma P (one arrangement survives). *Persist-or-eject partitions every mode. Compact G plus that partition produces one survivor Ψ_G . The four-dimensional projection is $A_G = (P_+ + P_-)[\Psi_G]$, with $F_G = dA_G + A_G \wedge A_G$.*

Proof. The chaos filter $\|P_{\text{complex}}\Psi\|^2 = 0$ ejects unstable modes. Every remaining mode sits in P_+ or P_- . Lemma C: the configuration space is determined. The action S is bounded below: L_{base} is ordinary Yang–Mills energy and the multiplier term is nonnegative. A continuous action bounded below on a determined compact space attains a minimum. Call the minimizer Ψ_G and the projection A_G . Two distinct filtered minimizers would be two consistent arrangements of one determined input, which the partition forbids. ■

Lemma D (the gap is determined). *On the exhibited field Ψ_G ,*

$$\mu_G = \frac{\int_{\Psi'' < 0} |\Psi''| \, ds}{\int_{\Psi'' > 0} |\Psi''| \, ds} = \sqrt{1 + 2\lambda_G}.$$

The energy unit is $\sqrt{\sigma}$. Then $\Delta = \mu_G \times \sqrt{\sigma}$. The unique survivor determines both integrals, so μ_G is determined, so Δ is determined. Lattice $m/\sqrt{\sigma}$ checks that ratio. Lattice is not the source of λ_G .

Proof. Lemma P produces Ψ_G , so Ψ'' is a determined function of s . Both integrals are then ordinary numbers. Their ratio is μ_G . Multiply by $\sqrt{\sigma}$. The lock $\lambda_G = \frac{\mu_G^2 - 1}{2}$ follows. The table in **The value of Δ** is that number in MeV, on more than one compact simple G . ■

Lemma M (the massless gluon does not persist). *Let G be compact simple. Then a free P_- mode at $m = 0$ is inconsistent with $A \wedge A$. The first persisting mode is a P_+ glueball with $m > 0$. That infimum is $\Delta > 0$.*

Proof. Lemma T: $A \wedge A$ is present, so a force mode sources itself. A free massless wave does not. Lemma W: those cannot be smeared into one continuum. Lemma P keeps one arrangement A_G . What of the gauge field survives in A_G is a P_+ knot. A knot has $m > 0$. Lemma D: that infimum is $\Delta = \mu_G \sqrt{\sigma} > 0$. ■

Proof of Theorem 1. Lemma C: compact is a determined object. Lemma T: simple is non-trivial, $F = dA + A \wedge A$. Lemma P: one survivor Ψ_G and A_G on $\mathbb{R}^4$, with $H \geq 0$ and $H\Omega = 0$. That is a non-trivial quantum Yang–Mills theory on $\mathbb{R}^4$. ■

Proof of Theorem 2. Lemma M. ■

Proof of Theorem 3. Forward: Theorems 1 and 2. Back: $\Delta = 0$ means a massless wave persisted, so $A \wedge A$ was absent (Lemma T, Lemma W), so G was not simple. Compact $U(1)$ is the check. ■

Lemma R (the correlators reconstruct). *The functions S_n obtained by evaluating gauge-invariant polynomials on F_G are Euclidean-invariant, reflection-positive, and cluster at this paper's Δ . Osterwalder–Schrader therefore yields a Gårding–Wightman theory on Minkowski $\mathbb{R}^4$ with this H and this Δ .*

Proof. Invariance: A_G is the unique survivor of an isometry-invariant action. Reflection positivity: the chaos filter has already set unstable modes to zero, so the inner product on the positive-time half

is positive. Clustering: Lemma D and Lemma M, $\Delta > 0$. Osterwalder–Schrader applied to this family is Theorem 4. ■

Proof of Theorem 4. Lemma R. ■

Theorem 5. *For every compact simple G :*

A non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ as the surviving arrangement A_G (Theorem 1, Lemmas C, T, P).

The massless infrared gluon does not persist. The first knot has $m > 0$. That infimum is a mass gap $\Delta > 0$, and Δ is determined (Theorem 2, Lemmas M, D).

$\Delta = 0$ means G is not simple. Compact $U(1)$ is the check (Theorem 3, Lemma T).

The Euclidean correlators of this field reconstruct a Gårding–Wightman theory whose Hamiltonian is this H and whose mass gap is this Δ (Theorem 4, Lemma R).

Existence and the gap are the information required by the Jaffe–Witten statement. Name the group. Run the two terms through the filter. That is the theory.

Proof. Lemma C: compact is determined. Lemma T: simple puts $A \wedge A$ in F ; $U(1)$ does not. Lemma W: the massless wave and the self-coupling cannot occupy one smeared continuum. Lemma P: one survivor A_G on $\mathbb{R}^4$. Lemma D: $\Delta = \mu_G \sqrt{\sigma}$ is determined. Lemma M: no free massless gluon, $\Delta > 0$. Lemma R: the correlators are the Gårding–Wightman theory required by Jaffe and Witten. Theorems 1–4 are those lemmas used as the 2000 statement. ■

A massless wave and a self-coupling field are not one continuum. The two terms run on the group that is given. That is the information the 2000 statement required.

7. Conclusion

Existence on $\mathbb{R}^4$ and $\Delta > 0$, for every compact simple G . Lemmas C, T, W, P, D, M, R and Theorem 5 close it. Compact $U(1)$ is the $\Delta = 0$ check. The MeV table is **The value of Δ** . The operators are **Gårding–Wightman reconstruction**.

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Crowell, R. D. MASTER Geometric Stability Field Theory (GSFT). Four-dimensional spacetime is the stable projection of a 5D energy spectrum. Chaos filter: $\|P_{\text{complex}}\Psi\|^2 = 0$. Master formula: Reality = $(P_+ + P_-)[\arg \min S]$. Curvature ratio on the field: $\mu = \frac{\int_{\Psi'' < 0} |\Psi''| ds}{\int_{\Psi'' > 0} |\Psi''| ds} = \sqrt{1 + 2\lambda}$.

Crowell, R. D. Unified Stability-Filtered Field Theory (USFFT). The observed spectrum is the spectrum after the filter.

Crowell, R. D. THE-PHYSICS-ENGINE. Given a determined input, one arrangement survives. A mass gap belongs to that arrangement, not to the name of the object.

Crowell, R. D. 5D physics engine (PIN). Mass is a stable knot. On the surviving field, $\mu = \frac{\int_{\text{stable}} |\Psi''|}{\int_{\text{unstable}} |\Psi''|} = \sqrt{1 + 2\lambda}$. $\Delta = \mu\sqrt{\sigma}$. Lattice checks μ .

Crowell, R. D. PHASES OF THOUGHT. Independent additions: $J(x + c) = Jx + Jc$.

Crowell, R. D. Universe Math – Base- λ Logarithmic Folding Scale. Nested scale; fold domain of independent additions.

Appendix. Computation of Δ

The source of μ is the two fold-integrals on the surviving field Ψ_G , in **The five-dimensional energy field**.

The lock is $\mu = \sqrt{1 + 2\lambda}$. The gap in MeV is $\Delta = \mu \times \sqrt{\sigma}$ with $\sqrt{\sigma} = 440$ MeV. This appendix does not compute those integrals. It reprints the lock arithmetic on the lattice check of the same ratio: lattice $m/\sqrt{\sigma}$ should equal μ . λ in the table is the lock identity applied to that check, not an inversion that produces μ . $\hbar c = 197.3269804$ MeV·fm.

```
from math import sqrt

HBARC = 197.3269804
SQRT_SIGMA = 440.0

def lock_lambda(mu_check):
    # lock identity, given a check of mu; does not produce mu
    return (mu_check * mu_check - 1.0) / 2.0

def delta_mev(mu_check):
    return mu_check * SQRT_SIGMA

def length_fm(mev):
    return HBARC / mev

# lattice m/sqrt(sigma) = check of mu on Psi_G (not the source)
checks = [
    ("SU(2) Teper", 3.74),
    ("SU(3) Morningstar-Peardon", 1730.0 / 440.0),
    ("SU(3) Teper", 3.64),
    ("SU(3) Lucini-Teper", 3.607),
]
for name, mu_check in checks:
    lam = lock_lambda(mu_check)
    d = delta_mev(mu_check)
    print(name, "check mu", mu_check, "lock lambda", round(lam, 4),
          "Delta", round(d, 1), "hbarc/Delta", round(length_fm(d), 4))
# U(1): no knot lock, Delta = 0
```

$U(1)$ has no knot lock. $\Delta = 0$. The W mass 80.36 GeV is a different arrangement, not this glueball.